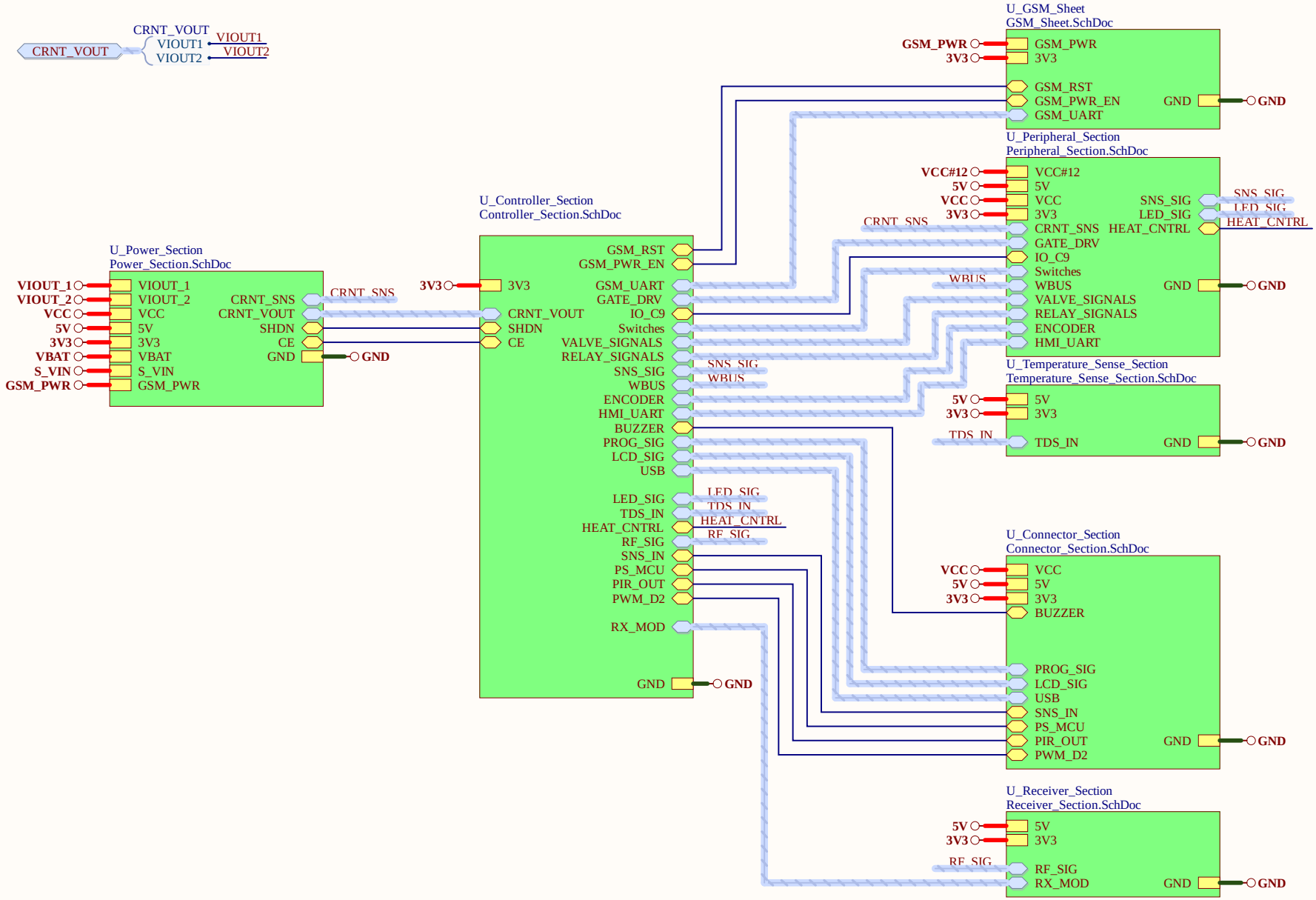
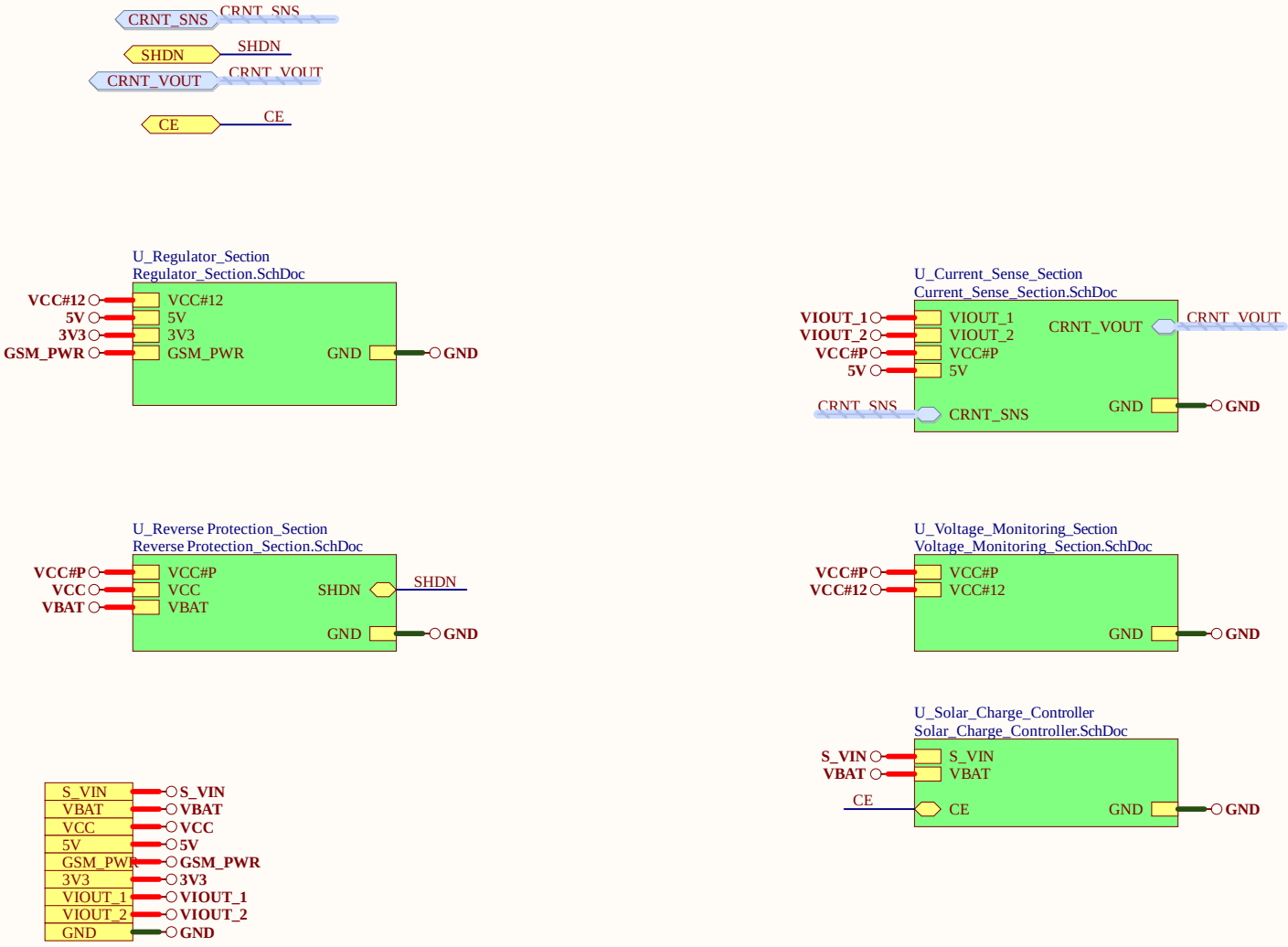
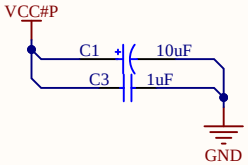

CRNT_VOUT
VIOUT1
VIOUT2
U_Power_Section
Power_Section.SchDoc
VIOUT_1
VIOUT_2
VCC
5V
3V3
VBAT
S_VIN
GSM_PWR
CRNT_SNS
CRNT_VOUT
SHDN
CE
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
GSM_RST
GSM_PWR_EN
GSM_UART
GATE_DRV
IO_C9
Switches
VALVE_SIGNALS
RELAY_SIGNALS
SNS_SIG
WBUS
ENCODER
HMI_UART
BUZZER
PROG_SIG
LCD_SIG
USB
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
SNS_IN
PS_MCU
PIR_OUT
PWM_D2
RX_MOD
U_GSM_Sheet
GSM_Sheet.SchDoc
U_Peripheral_Section
Peripheral_Section.SchDoc
VCC#12
U_Temperature_Sense_Section
Temperature_Sense_Section.SchDoc
U_Connector_Section
Connector_Section.SchDoc
U_Receiver_Section
Receiver_Section.SchDoc

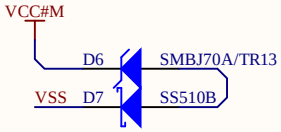

CRNT_SNS
CRNT_SNS
SHDN
SHDN
CRNT_VOUT
CRNT_VOUT
CE
CE
U_Regulator_Section
Regulator_Section.SchDoc
VCC#12
5V
3V3
GSM_PWR
VCC#12
5V
3V3
GSM_PWR
GND
GND
U_Current_Sense_Section
Current_Sense_Section.SchDoc
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_SNS
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_VOUT
CRNT_VOUT
CRNT_SNS
GND
GND
U_Reverse Protection_Section
Reverse Protection_Section.SchDoc
VCC#P
VCC
VBAT
VCC#P
VCC
VBAT
SHDN
SHDN
GND
GND
U_Voltage_Monitoring_Section
Voltage_Monitoring_Section.SchDoc
VCC#P
VCC#12
VCC#P
VCC#12
GND
GND
U_Solar_Charge_Controller
Solar_Charge_Controller.SchDoc
S_VIN
VBAT
CE
S_VIN
VBAT
CE
GND
GND
S_VIN
VBAT
VCC
5V
GSM_PWR
3V3
VIOUT_1
VIOUT_2
GND
S_VIN
VBAT
VCC
5V
GSM_PWR
3V3
VIOUT_1
VIOUT_2
GND

SHDN — SHDN

DECOUPLING CAPACITOR

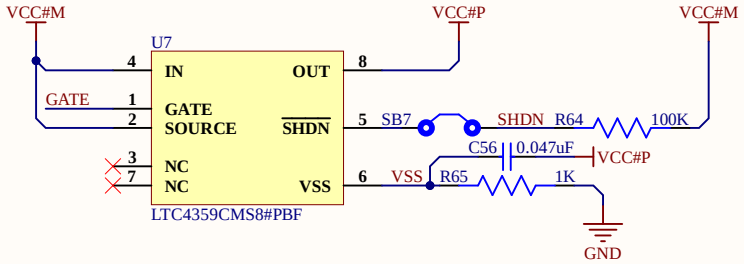


INPUT DIODES

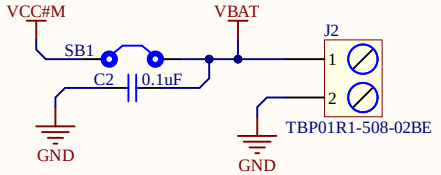


VBAT — VBAT
VCC — VCC
VCC#P — VCC#P
GND — GND

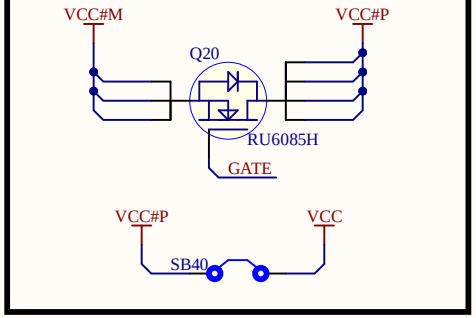
REVERSE POLARITY PROTECTION IC



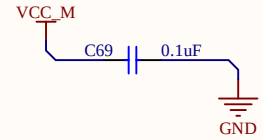
BATTERY / MAIN INPUT



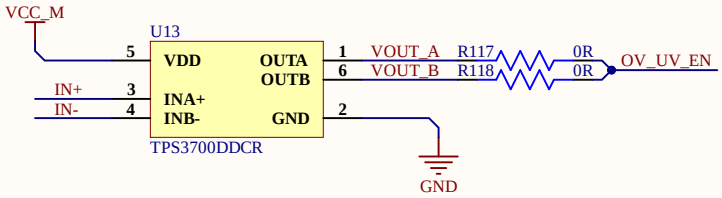
MOSFET



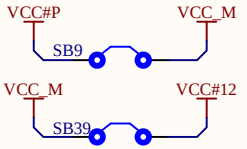
DECOUPLING CAPACITOR



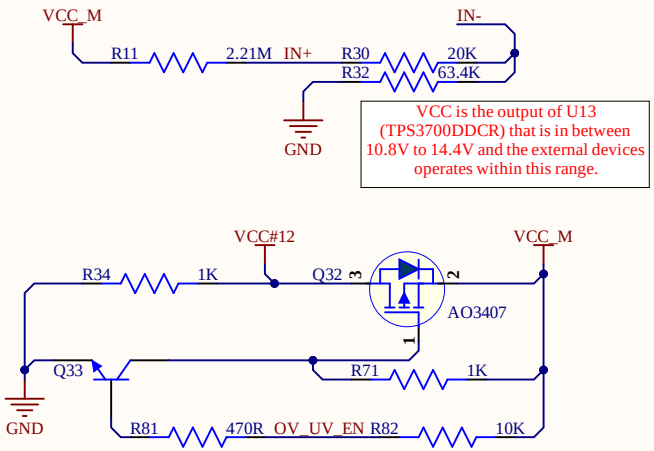
UNDER AND OVER VOLTAGE PROTECTION IC

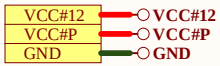


VOLTAGE TO MONITOR

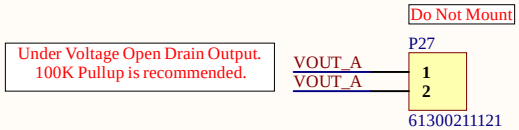


SUPPORTING CIRCUITRY

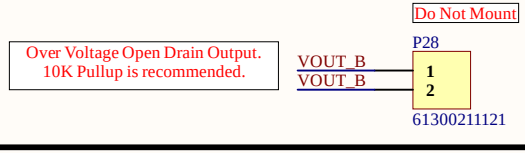


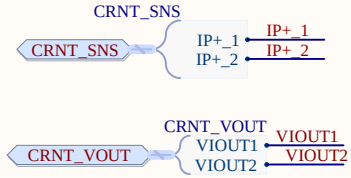


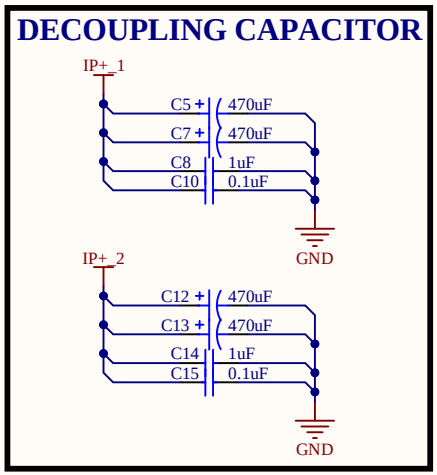
UNDER VOLTAGE OUTPUT

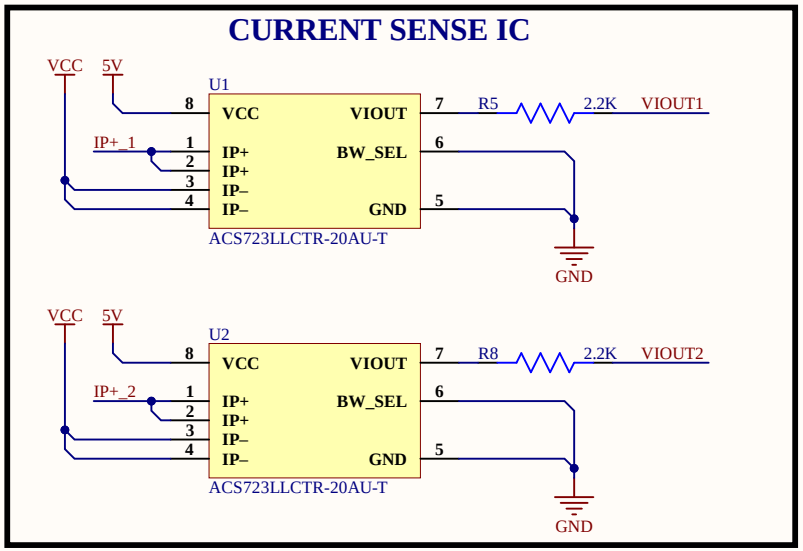


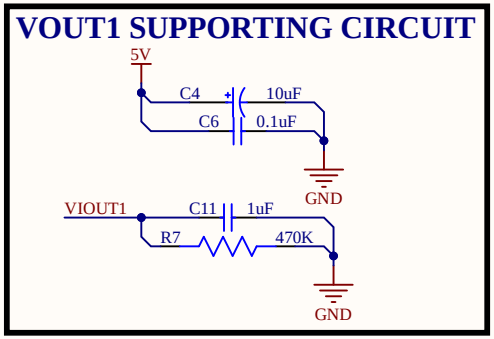
OVER VOLTAGE OUTPUT

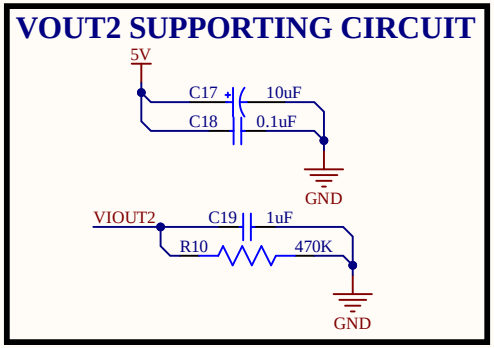




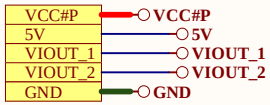


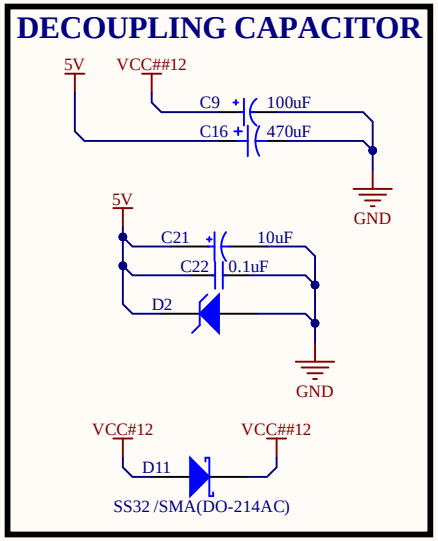


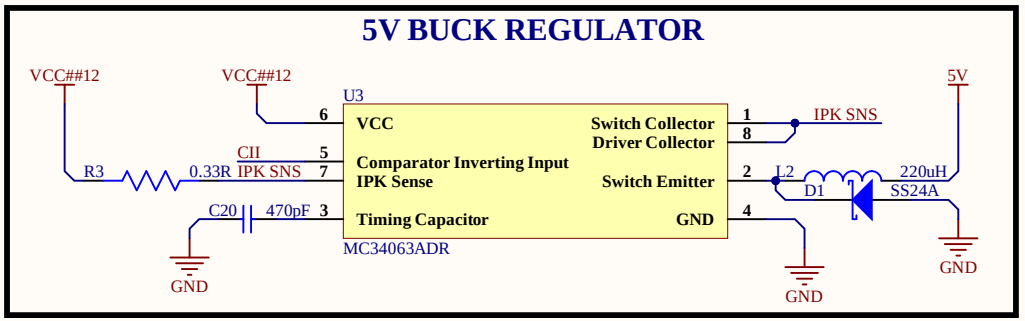


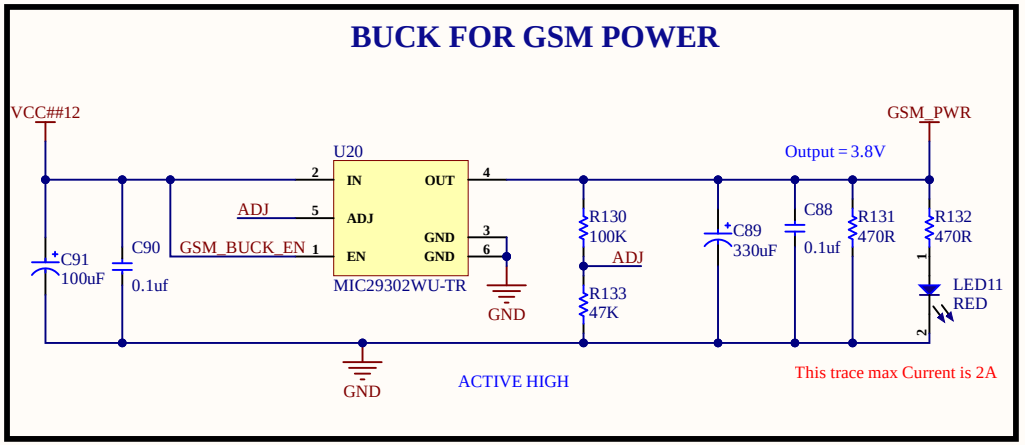


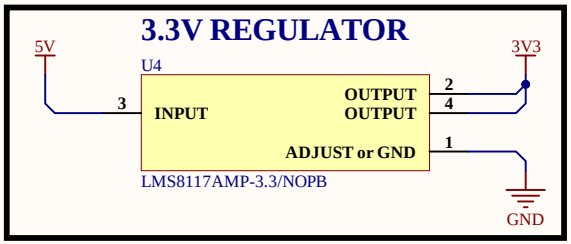
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

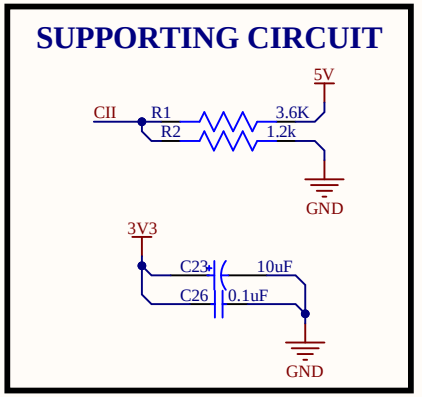




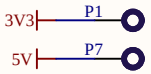




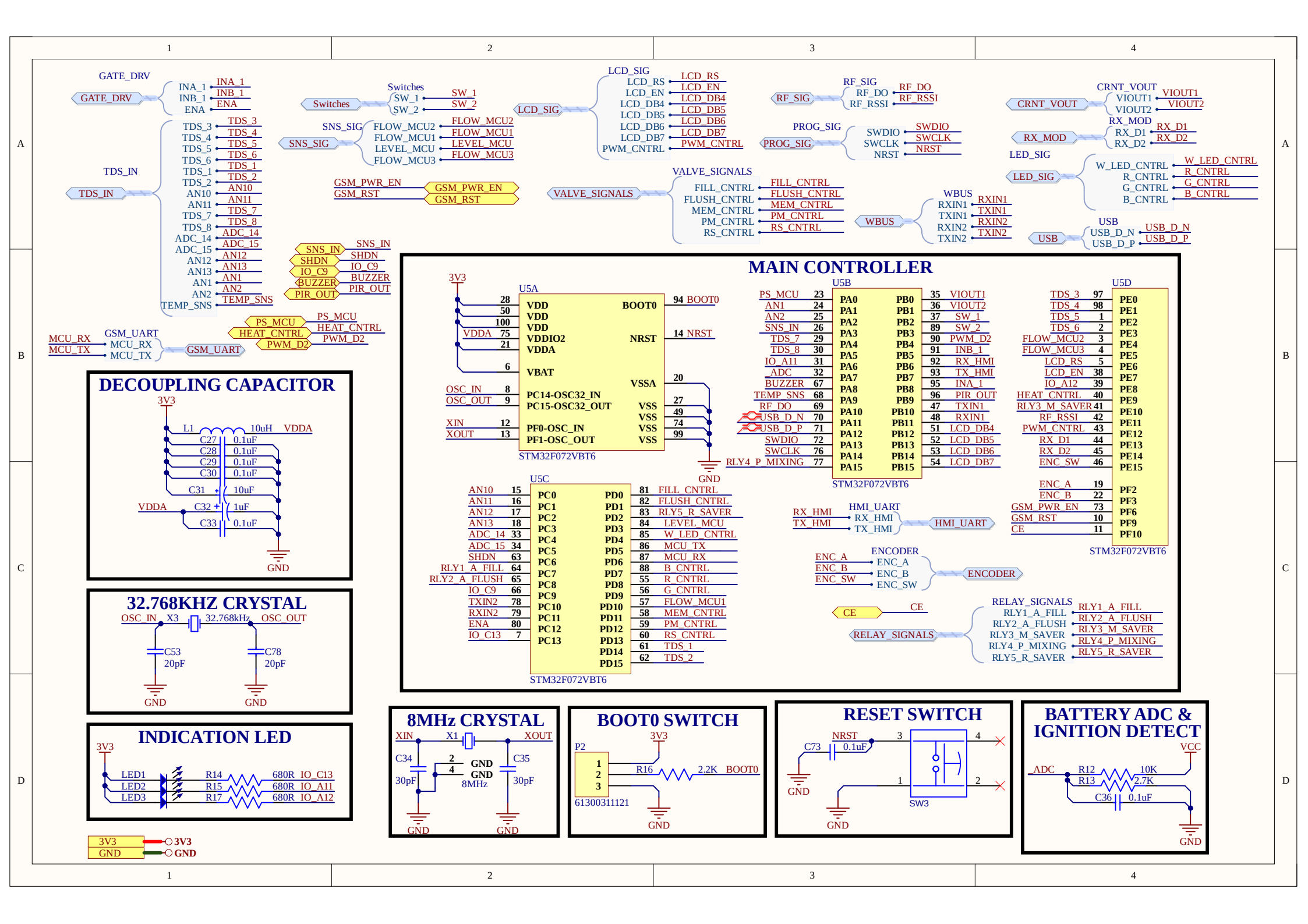


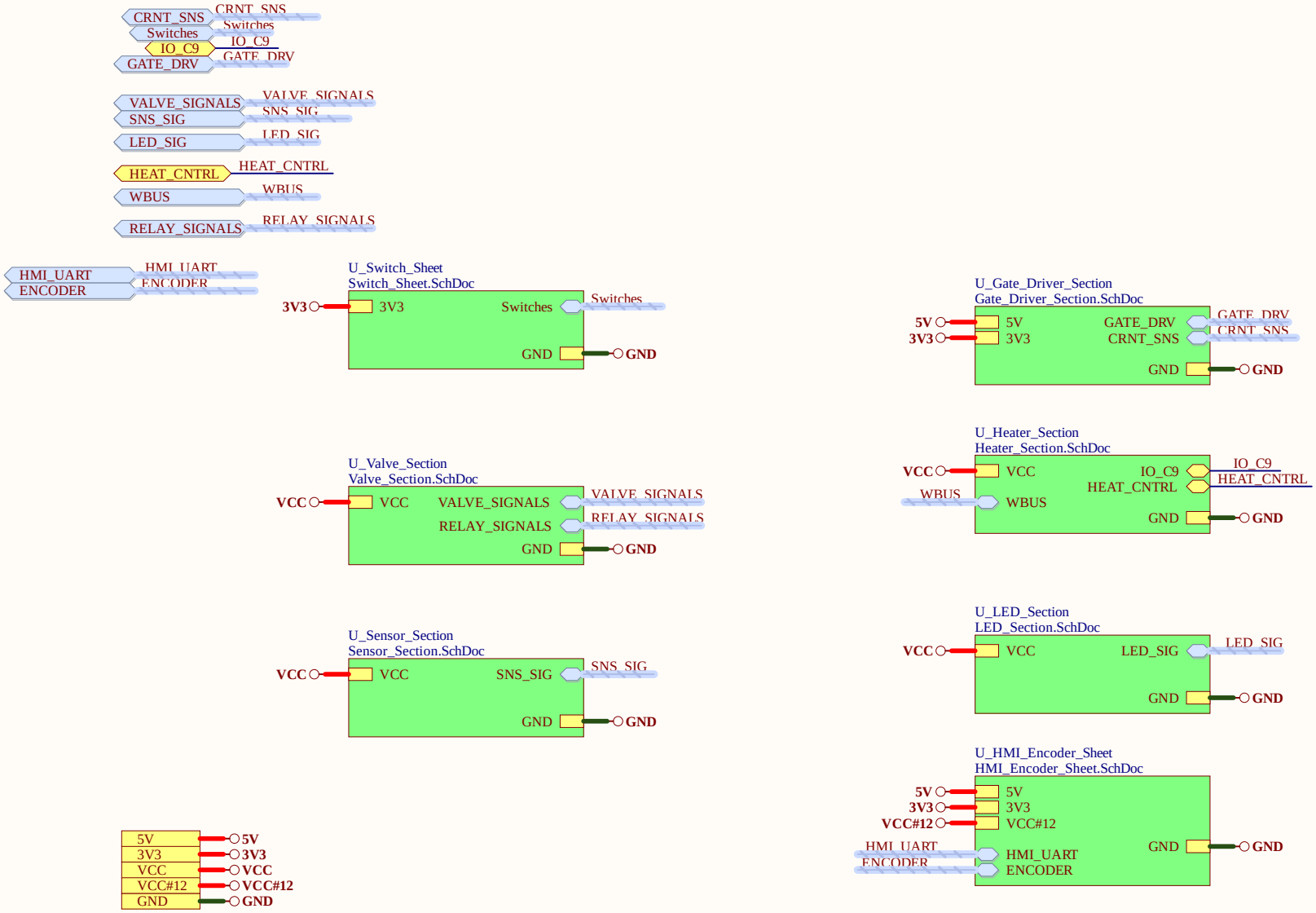


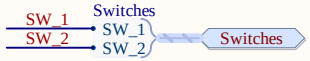
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

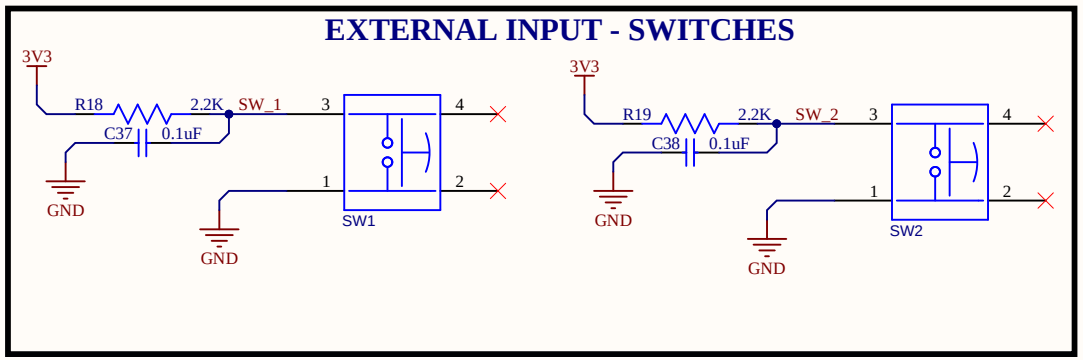


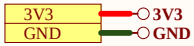
VCC#12	VCC#12
5V	5V
GSM_PWR	GSM_PWR
3V3	3V3
GND	GND

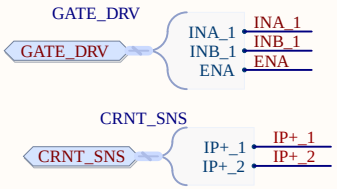

GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
TEMP_SNS
Switches
SW_1
SW_2
SNS_SIG
FLOW_MCU2
FLOW_MCU1
LEVEL_MCU
FLOW_MCU3
GSM_PWR_EN
GSM_RST
LCD_SIG
LCD_RS
LCD_EN
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
PWM_CNTRL
RF_SIG
RF_DO
RF_RSSI
PROG_SIG
SWDIO
SWCLK
NRST
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
CRNT_VOUT
VIOUT1
VIOUT2
RX_MOD
RX_D1
RX_D2
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
USB
USB_D_N
USB_D_P
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
PWM_D2
MCU_RX
MCU_TX
GSM_UART
MAIN CONTROLLER
U5A
STM32F072VBT6
U5B
U5C
U5D
DECOUPLING CAPACITOR
3V3
L1
10uH
VDDA
C27
C28
C29
C30
C31
C32
C33
0.1uF
10uF
1uF
GND
32.768KHZ CRYSTAL
OSC_IN
X3
32.768kHz
OSC_OUT
C53
C78
20pF
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R
IO_C13
IO_A11
IO_A12
8MHz CRYSTAL
XIN
X1
XOUT
C34
C35
30pF
8MHz
BOOT0 SWITCH
P2
R16
2.2K
BOOT0
61300311121
RESET SWITCH
NRST
C73
SW3
BATTERY ADC & IGNITION DETECT
VCC
ADC
R12
R13
10K
2.7K
C36


CRNT_SNS
Switches
IO_C9
GATE_DRV
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
WBUS
RELAY_SIGNALS
HMI_UART
ENCODER
U_Switch_Sheet
Switch_Sheet.SchDoc
3V3
GND
U_Gate_Driver_Section
Gate_Driver_Section.SchDoc
5V
U_Valve_Section
Valve_Section.SchDoc
VCC
U_Heater_Section
Heater_Section.SchDoc
U_Sensor_Section
Sensor_Section.SchDoc
U_LED_Section
LED_Section.SchDoc
U_HMI_Encoder_Sheet
HMI_Encoder_Sheet.SchDoc
VCC#12

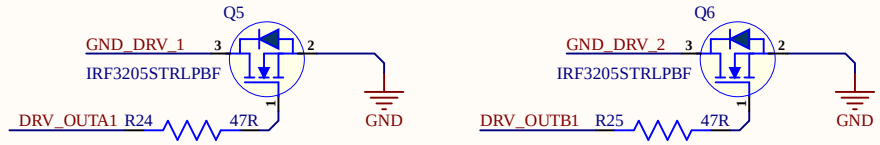

SW_1
SW_2
Switches
SW_1
SW_2
Switches


EXTERNAL INPUT - SWITCHES
3V3
R18
2.2K
SW_1
C37
0.1uF
GND
3
4
1
2
SW1
GND
3V3
R19
2.2K
SW_2
C38
0.1uF
GND
3
4
1
2
SW2
GND

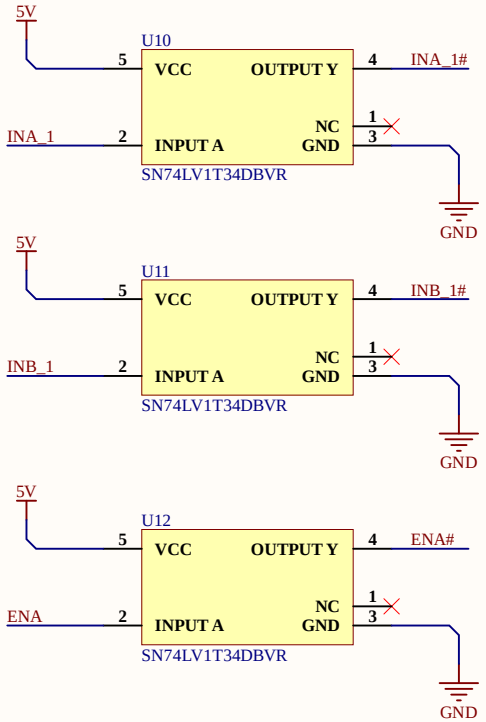

3V3
GND
3V3
GND



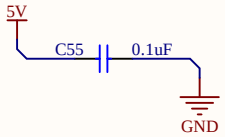
MOSFET DRIVERS



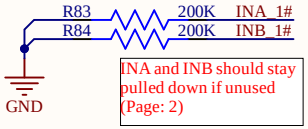
LEVEL TRANSLATOR



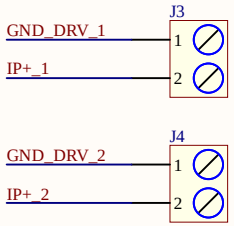
DECOUPLING CAPACITOR



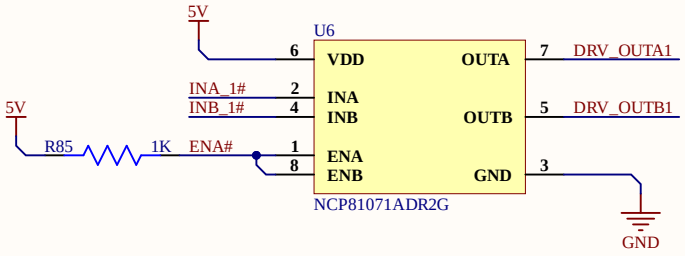
PULL DOWN RESISTOR

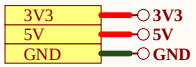


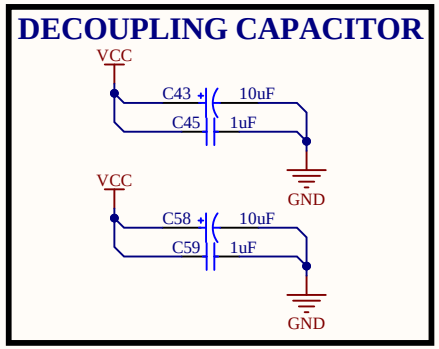
HEADERS FOR PUMP

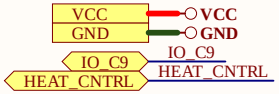


LOW SIDE PUMP DRIVER IC

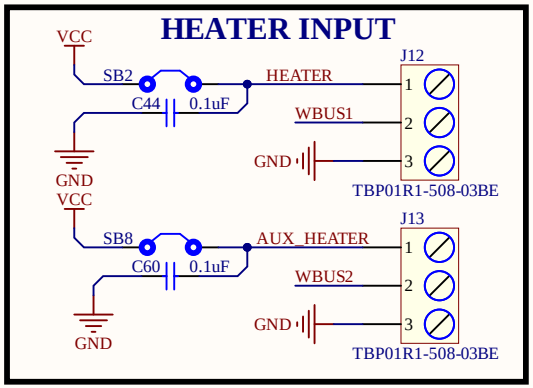


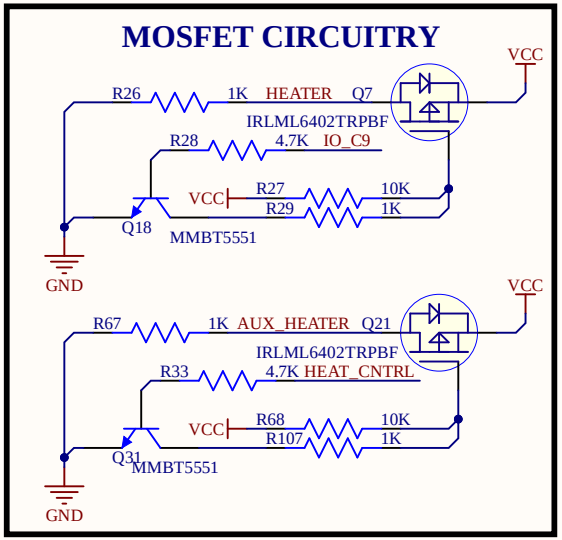


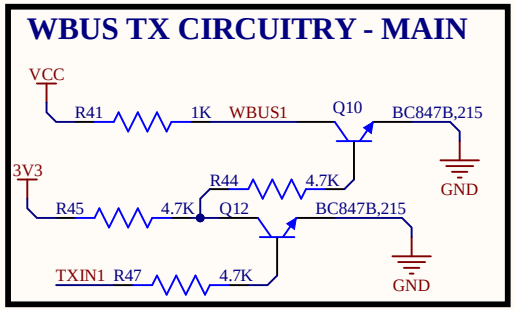

DECOUPLING CAPACITOR
VCC
C43
10uF
C45
1uF
GND
VCC
C58
10uF
C59
1uF
GND

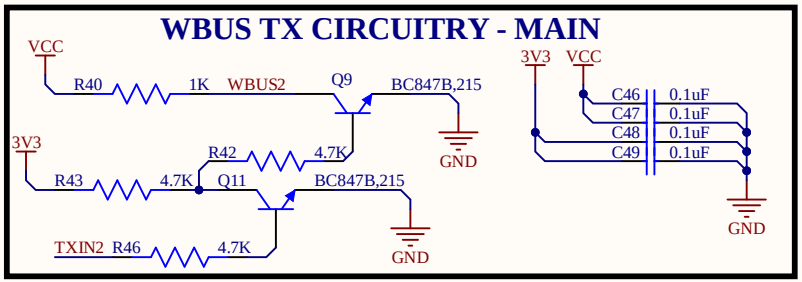

VCC
GND
IO_C9
HEAT_CNTRL
VCC
GND
IO_C9
HEAT_CNTRL

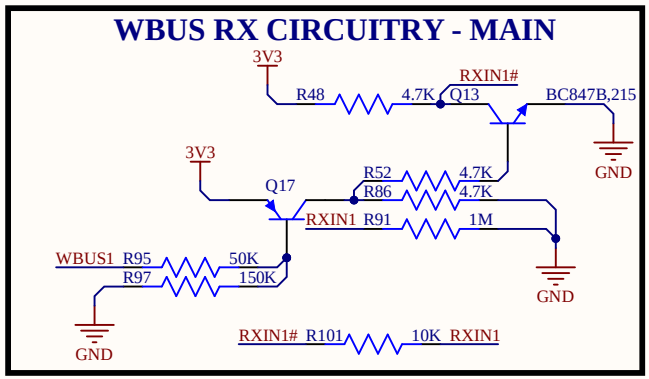

WBUS
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
RXIN1
TXIN1
RXIN2
TXIN2

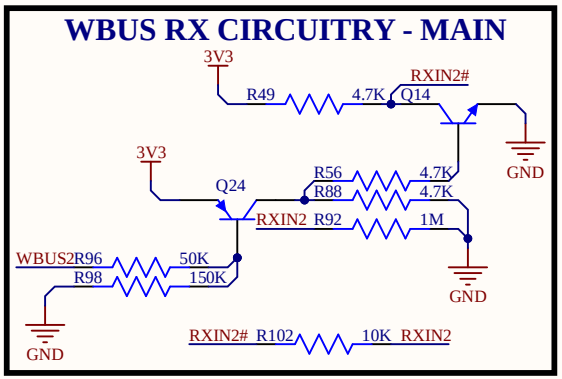

HEATER INPUT
VCC
SB2
C44
0.1uF
HEATER
J12
WBUS1
GND
TBP01R1-508-03BE
GND
VCC
SB8
C60
0.1uF
AUX_HEATER
J13
WBUS2
GND
TBP01R1-508-03BE
GND

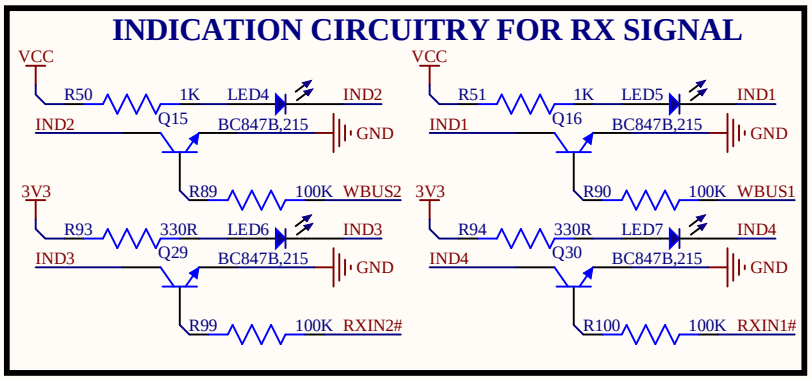

MOSFET CIRCUITRY
VCC
R26
1K
HEATER
Q7
IRLML6402TRPBF
R28
4.7K
IO_C9
VCC
R27
10K
R29
1K
Q18
MMBT5551
GND
VCC
R67
1K
AUX_HEATER
Q21
IRLML6402TRPBF
R33
4.7K
HEAT_CNTRL
VCC
R68
10K
R107
1K
Q31
MMBT5551
GND


WBUS TX CIRCUITRY - MAIN
VCC
R41
1K
WBUS1
Q10
BC847B,215
GND
3V3
R44
4.7K
R45
4.7K
Q12
BC847B,215
GND
TXIN1
R47
4.7K
GND

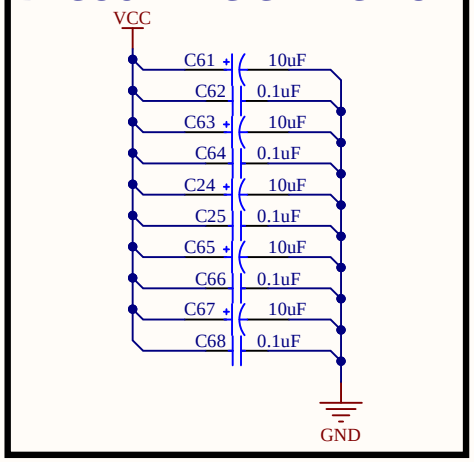

WBUS TX CIRCUITRY - MAIN
VCC
R40
1K
WBUS2
Q9
BC847B,215
GND
3V3
VCC
C46
0.1uF
C47
0.1uF
C48
0.1uF
C49
0.1uF
GND
3V3
R42
4.7K
R43
4.7K
Q11
BC847B,215
GND
TXIN2
R46
4.7K
GND


WBUS RX CIRCUITRY - MAIN
3V3
R48
4.7K
RXIN1#
Q13
BC847B,215
GND
3V3
Q17
R52
4.7K
R86
4.7K
RXIN1
R91
1M
WBUS1
R95
50K
R97
150K
GND
GND
RXIN1#
R101
10K
RXIN1

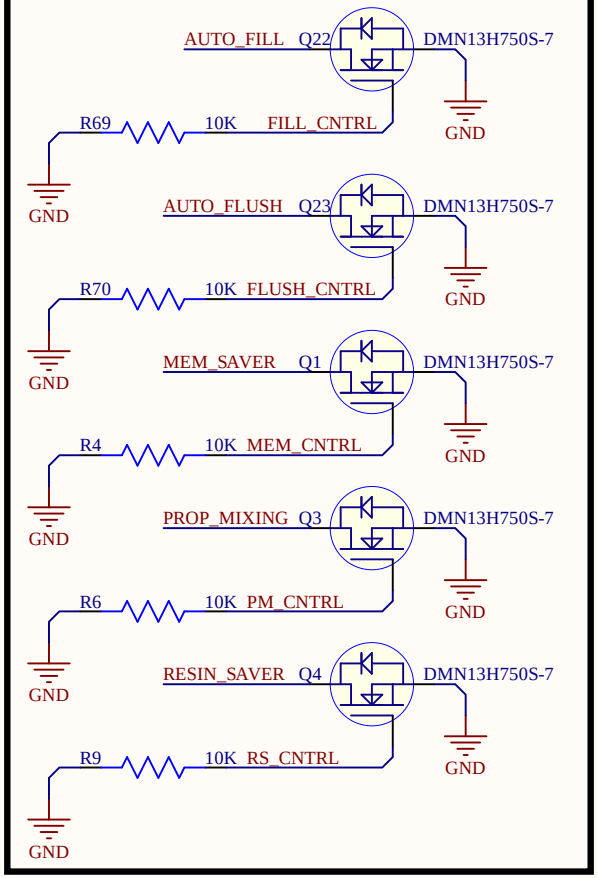

WBUS RX CIRCUITRY - MAIN
3V3
R49
4.7K
RXIN2#
Q14
BC847B,215
GND
3V3
Q24
R56
4.7K
R88
4.7K
RXIN2
R92
1M
WBUS2
R96
50K
R98
150K
GND
GND
RXIN2#
R102
10K
RXIN2


INDICATION CIRCUITRY FOR RX SIGNAL
VCC
R50
1K
LED4
IND2
IND2
Q15
BC847B,215
GND
VCC
R51
1K
LED5
IND1
IND1
Q16
BC847B,215
GND
3V3
R89
100K
WBUS2
3V3
R90
100K
WBUS1
R93
330R
LED6
IND3
IND3
Q29
BC847B,215
GND
R94
330R
LED7
IND4
IND4
Q30
BC847B,215
GND
R99
100K
RXIN2#
R100
100K
RXIN1#

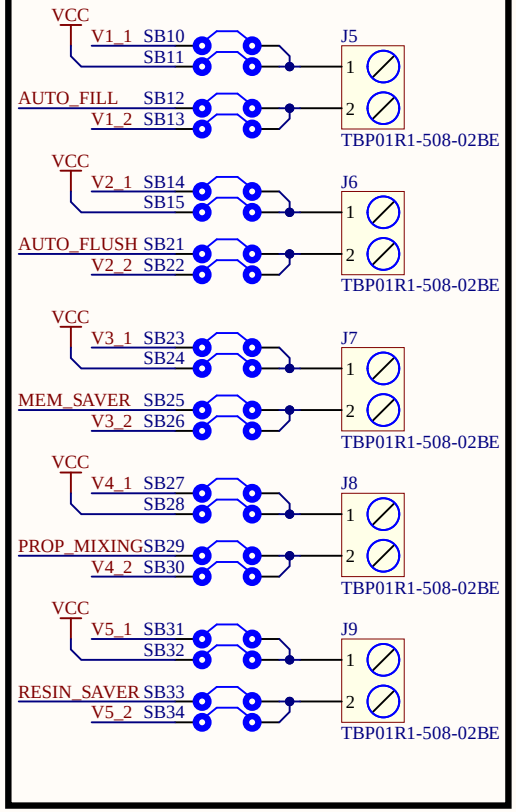
DECOUPLING CAPACITOR

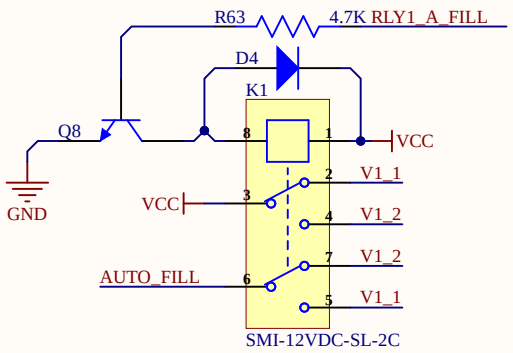


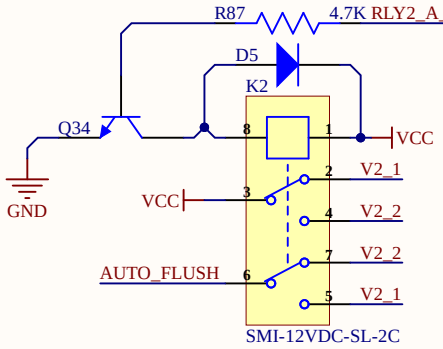
SOLENOID CONTROL CIRCUIT

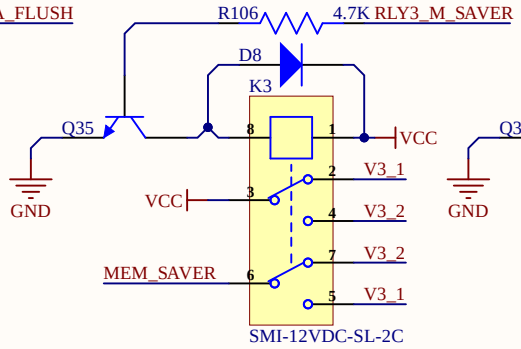


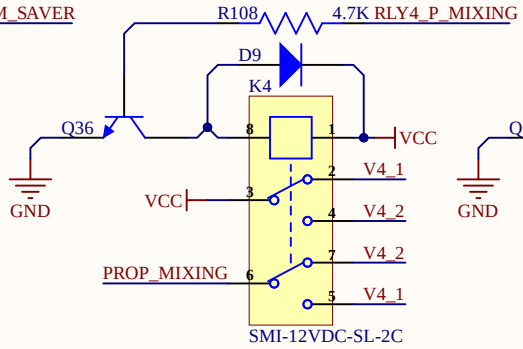
SOLENOID VALVE INPUTS

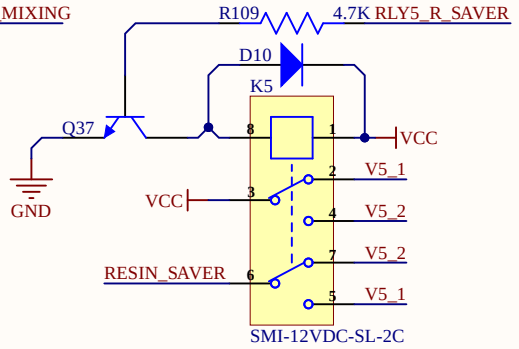


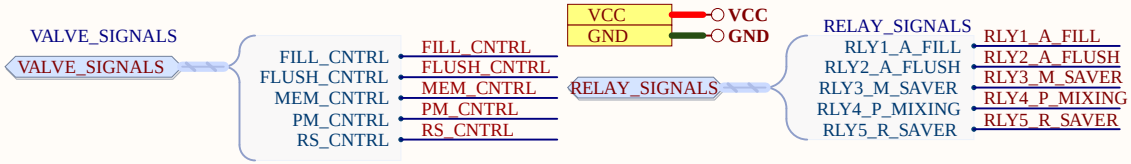




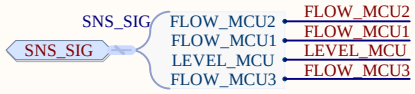




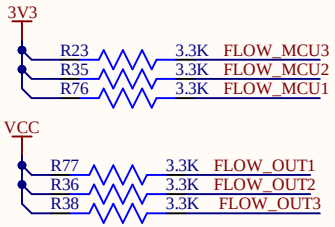




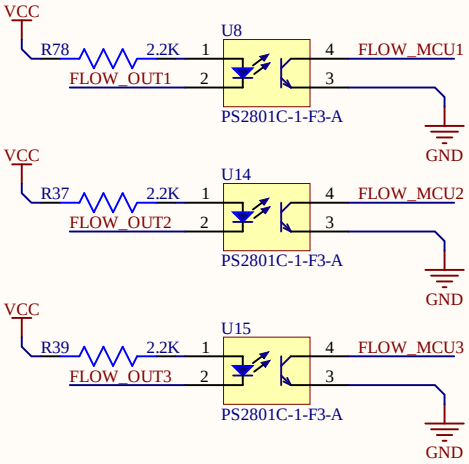
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PULL UP RESISTORS

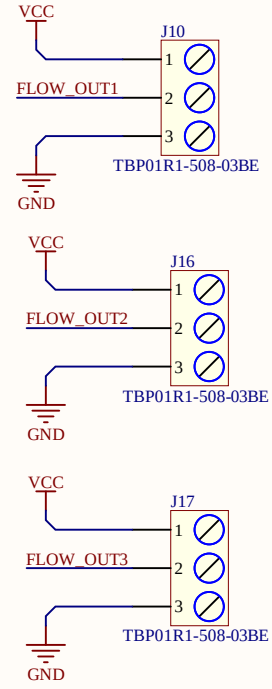


OPTO ISOLATION FOR FLOW METER

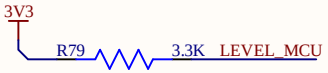


Forward voltage of diode=1.2V
With VCC=12V, IF=5.4mA
Flow meter datasheet recommends a pull up value between 3K-30K

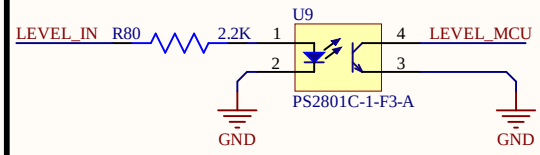
FLOW METER INPUT



SUPPORTING CIRCUIT

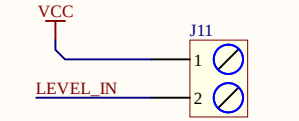


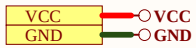
OPTO ISOLATION FOR FLOW METER



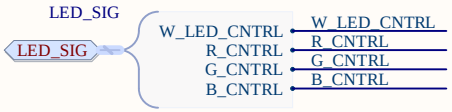
Forward voltage of diode=1.2V
With VCC=12V, IF=54mA

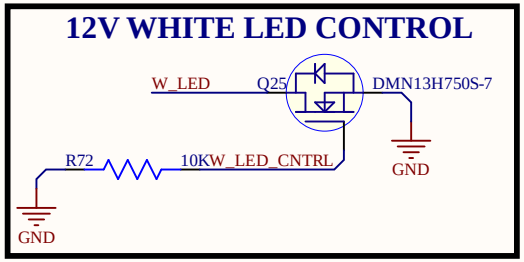
LEVEL SENSOR INPUT

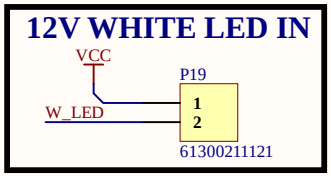


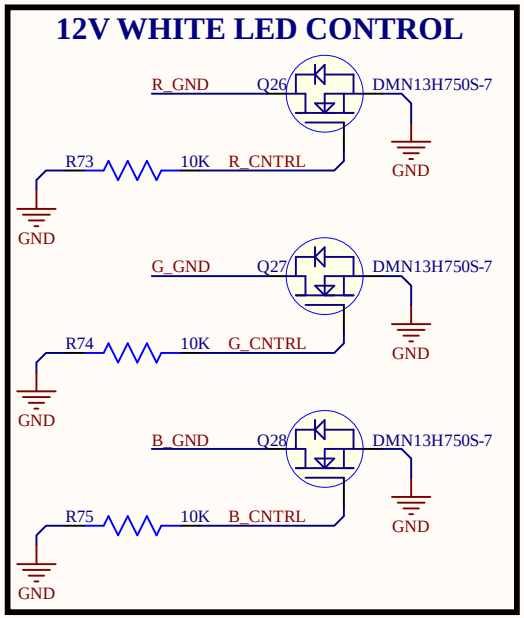


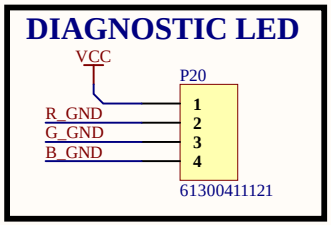
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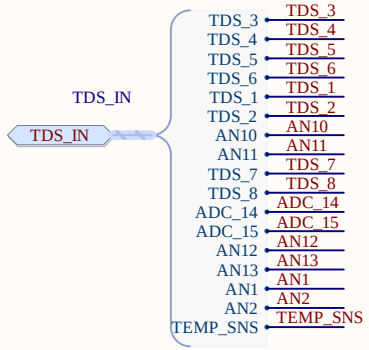




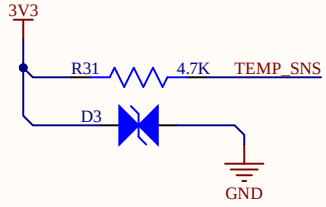




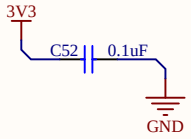
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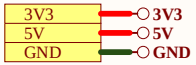


PULL UP RESISTORS

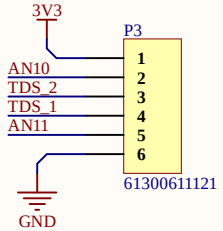


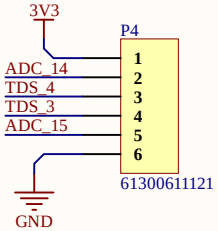
DECOUPLING CAP

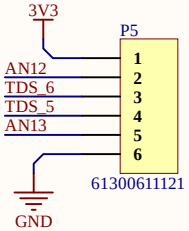


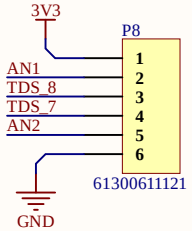


4 X TDS INPUTS



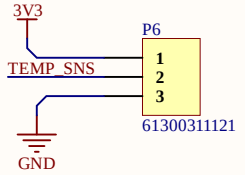


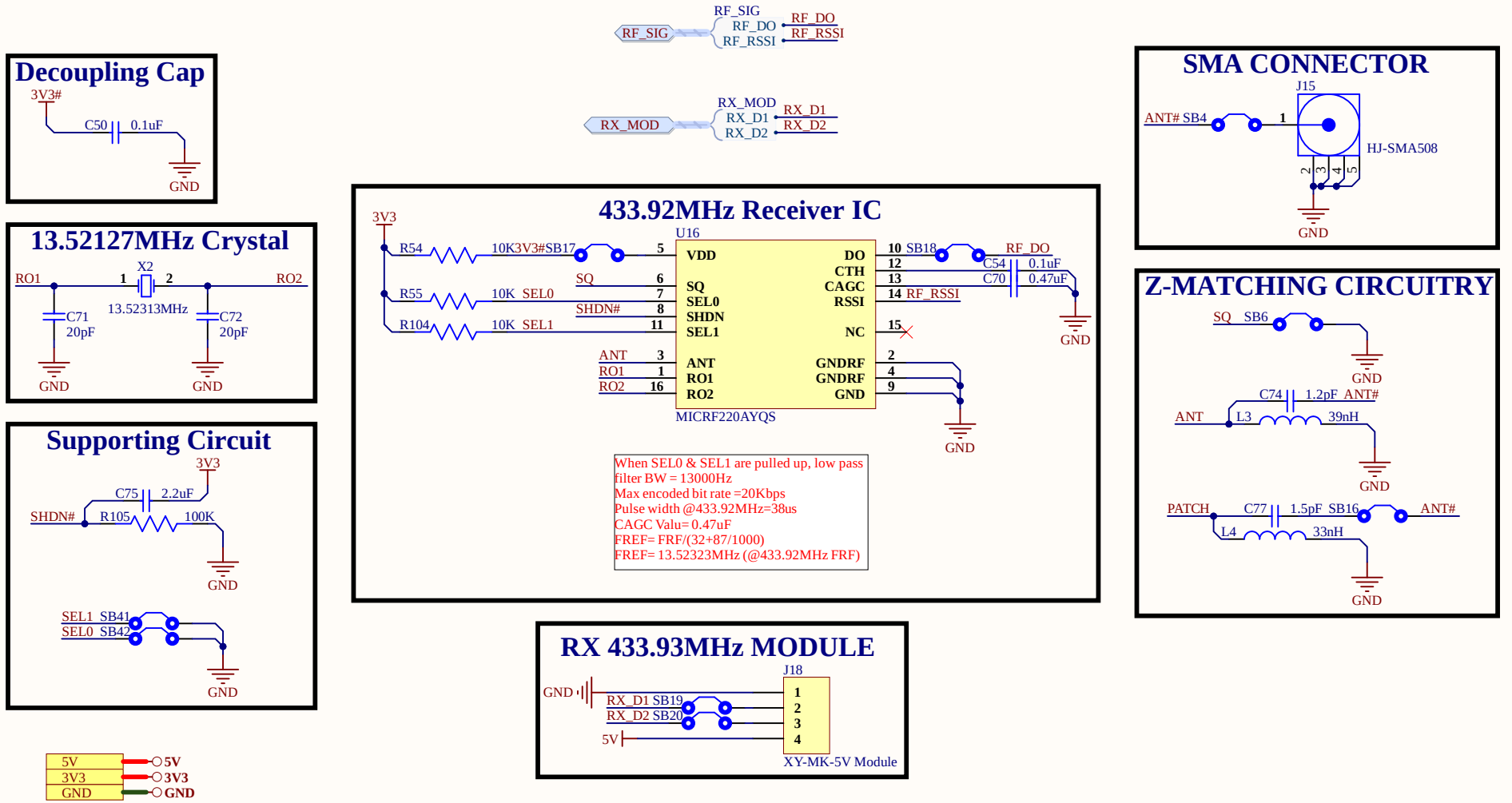


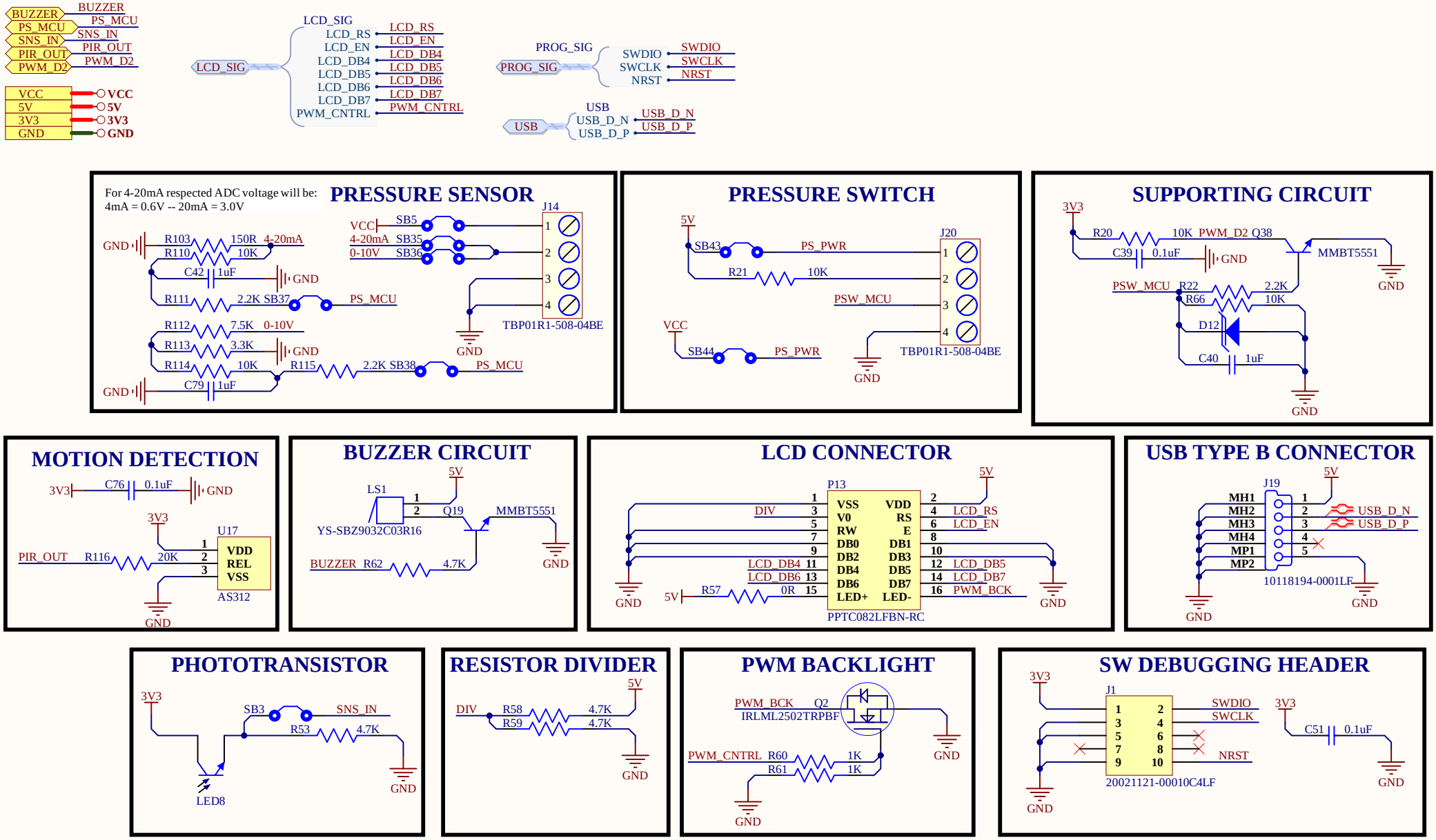


These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

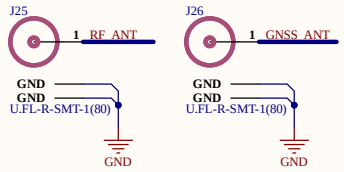
TEMP SENSOR DS18B20



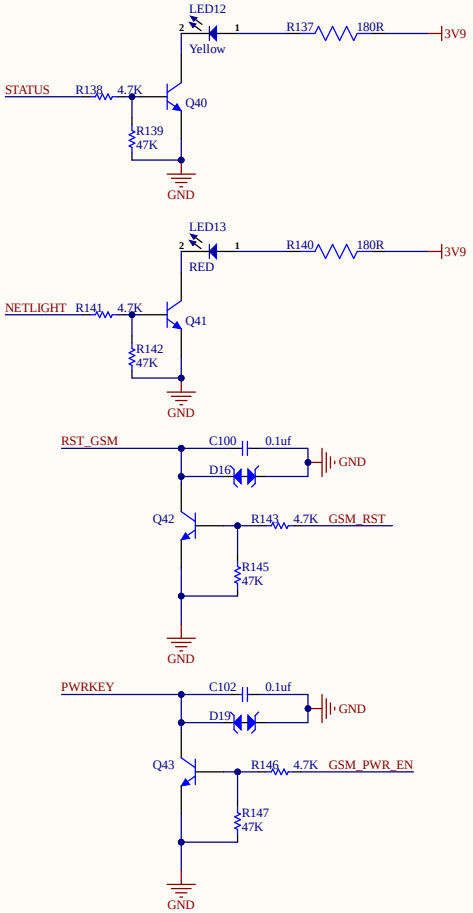

Decoupling Cap
3V3#
C50
0.1uF
GND
13.52127MHz Crystal
RO1
X2
1
2
RO2
13.52313MHz
C71
20pF
C72
20pF
GND
GND
Supporting Circuit
3V3
C75
2.2uF
SHDN#
R105
100K
GND
SEL1 SB41
SEL0 SB42
GND
5V
3V3
GND
5V
3V3
GND
RF_SIG
RF_SIG
RF_DO
RF_RSSI
RF_DO
RF_RSSI
RX_MOD
RX_MOD
RX_D1
RX_D2
RX_D1
RX_D2
433.92MHz Receiver IC
3V3
U16
R54
10K3V3#SB17
5
VDD
R55
10K SEL0
SQ
6
SQ
R104
10K SEL1
SHDN#
8
SEL0
SHDN
11
SEL1
ANT
3
ANT
RO1
1
RO1
RO2
16
RO2
DO
10
SB18
RF_DO
CTH
12
C54
0.1uF
CAGC
13
C70
0.47uF
RSSI
14
RF_RSSI
NC
15
GNDRF
2
GNDRF
4
GND
9
GND
GND
MICRF220AYQS
When SEL0 & SEL1 are pulled up, low pass
filter BW = 13000Hz
Max encoded bit rate =20Kbps
Pulse width @433.92MHz=38us
CAGC Valu= 0.47uF
FREF= FRF/(32+87/1000)
FREF= 13.52323MHz (@433.92MHz FRF)
RX 433.93MHz MODULE
J18
GND
RX_D1 SB19
RX_D2 SB20
5V
1
2
3
4
XY-MK-5V Module
SMA CONNECTOR
J15
ANT# SB4
1
HJ-SMA508
2
3
4
5
GND
Z-MATCHING CIRCUITRY
SQ SB6
GND
C74
1.2pF
ANT#
ANT
L3
39nH
GND
PATCH
C77
1.5pF
SB16
ANT#
L4
33nH
GND
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Title
V3_Board.PrjPcb
Size
A4
Number
1
Revision
1
Date:
8/31/2023
Sheet 16 of 17
File:
D:\Github\..\Receiver_Section.SchDoc
Drawn By:
Huzaifa Ali


BUZZER
PS_MCU
SNS_IN
PIR_OUT
PWM_D2
VCC
5V
3V3
GND
LCD_SIG
LCD_RS
LCD_EN
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
PWM_CNTRL
PROG_SIG
SWDIO
SWCLK
NRST
USB
USB_D_N
USB_D_P
PRESSURE SENSOR
For 4-20mA respected ADC voltage will be:
4mA = 0.6V -- 20mA = 3.0V
J14
TBP01R1-508-04BE
PRESSURE SWITCH
J20
SB43
SB44
PS_PWR
PSW_MCU
R21
10K
SUPPORTING CIRCUIT
MMBT5551
R20
10K_PWM_D2
Q38
C39
0.1uF
R22
2.2K
R66
D12
C40
MOTION DETECTION
U17
AS312
C76
R116
20K
BUZZER CIRCUIT
LS1
YS-SBZ9032C03R16
Q19
MMBT5551
R62
4.7K
LCD CONNECTOR
P13
PPTC082LFBN-RC
R57
0R
USB TYPE B CONNECTOR
J19
10118194-0001LF
PHOTOTRANSISTOR
SB3
R53
LED8
RESISTOR DIVIDER
R58
R59
PWM BACKLIGHT
Q2
IRLML2502TRPBF
R60
R61
1K
SW DEBUGGING HEADER
J1
20021121-00010C4LF
C51

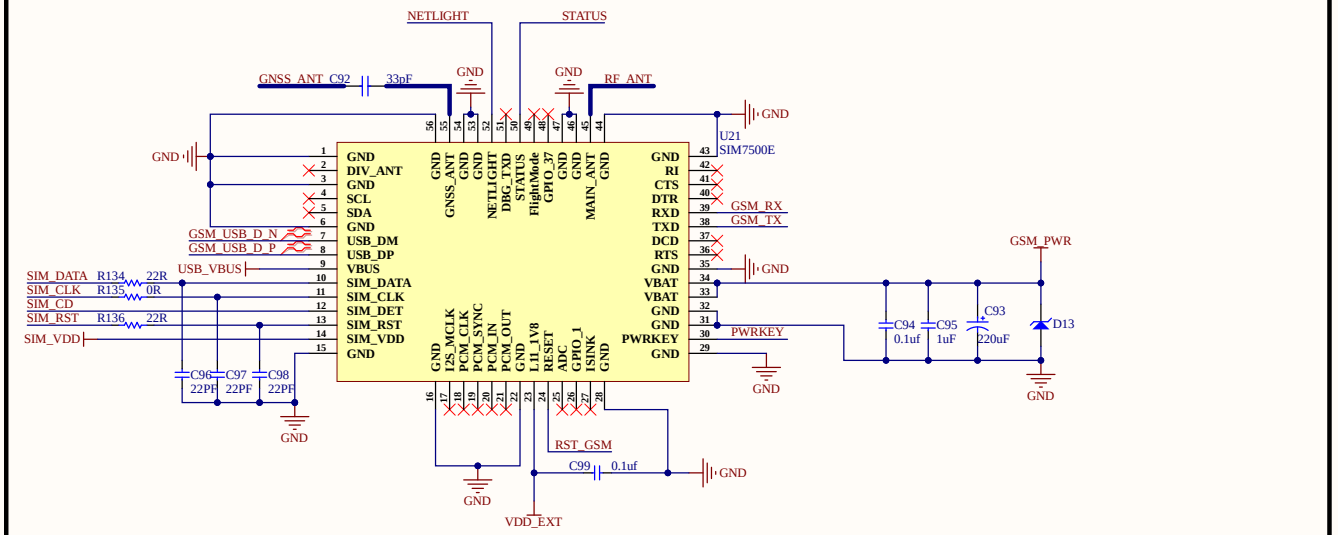
ANTENNA



GSM_COMMANDS



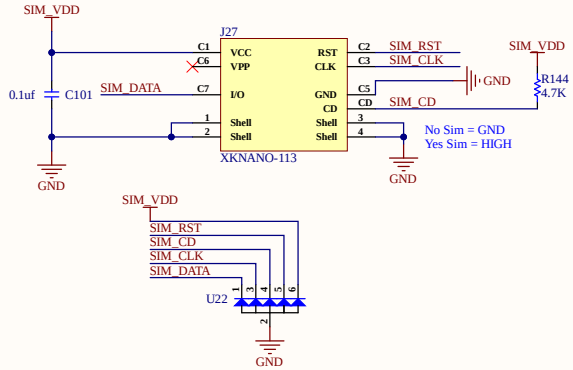
SIM7500E



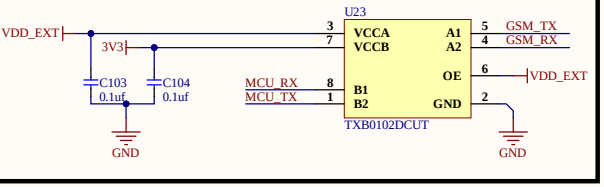
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

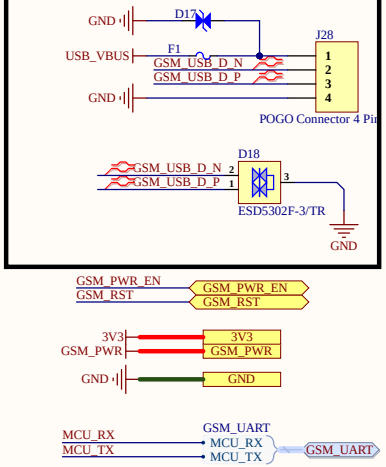
NANOSIM CONNECTOR



Level Shifter for GSM Serial

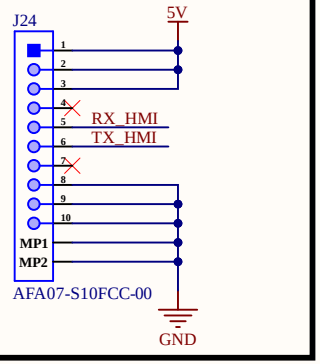


USB POGO for Updates

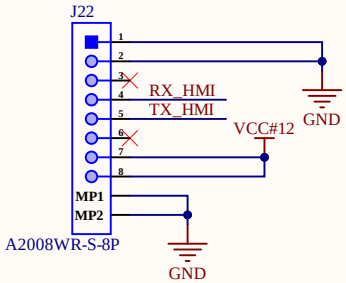


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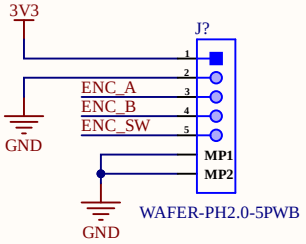
HMI INPUT DWIN

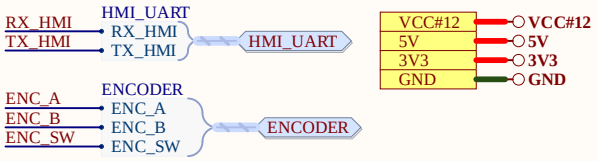


HMI INPUT STONE

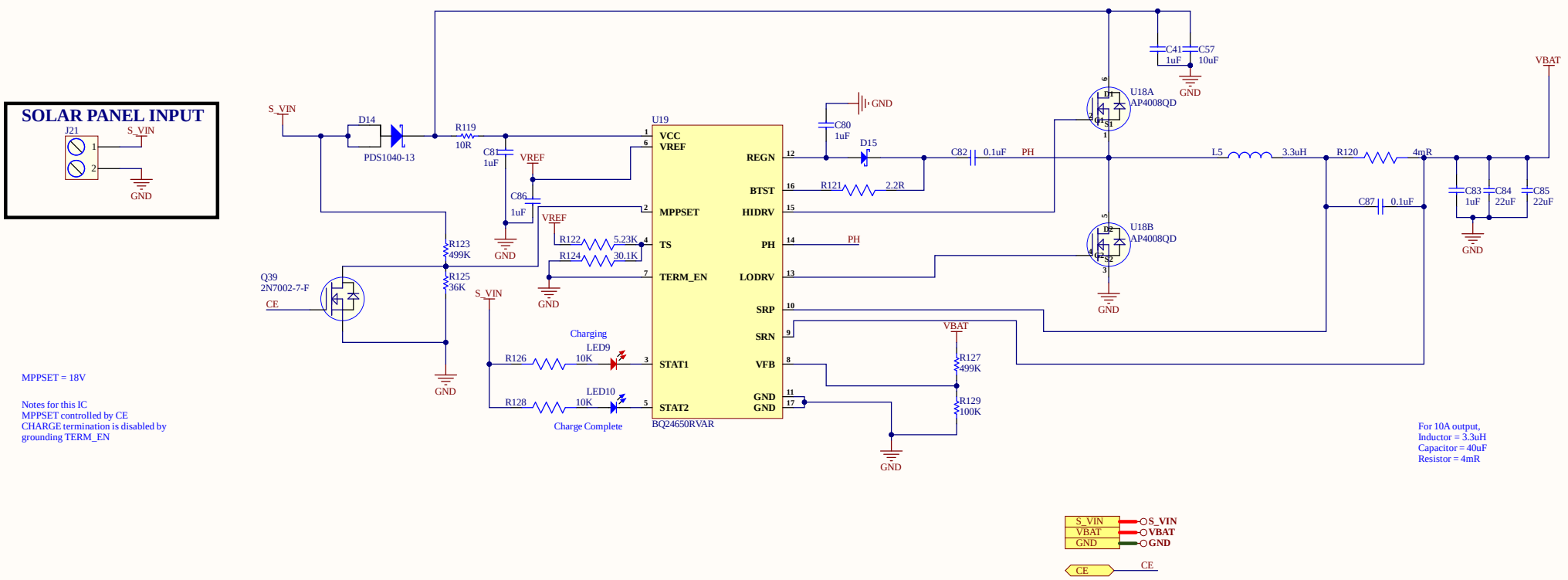


ENCODER INPUT





Title		
Size: A4	Number	Revision
Date: 8/31/2023		Sheet of
File: D:\Github\..\HMI_Encoder_Sheet.SchDoc		Drawn By:


SOLAR PANEL INPUT
J21
S_VIN
1
2
GND
S_VIN
D14
PDS1040-13
R119
10R
C81
1uF
VREF
C86
1uF
R123
499K
R125
36K
Q39
2N7002-7-F
CE
GND
R122
5.23K
R124
30.1K
U19
VCC
VREF
MPPSET
TS
TERM_EN
STAT1
STAT2
REGN
BTST
HIDRV
PH
LODRV
SRP
SRN
VFB
GND
GND
BQ24650RVAR
C80
1uF
D15
R121
2.2R
C82
0.1uF
PH
C41
1uF
C57
10uF
U18A
AP4008QD
U18B
AP4008QD
L5
3.3uH
R120
4mR
C87
0.1uF
C83
1uF
C84
22uF
C85
22uF
VBAT
Charging
LED9
10K
R126
LED10
10K
R128
Charge Complete
R127
499K
R129
100K
MPPSET = 18V
Notes for this IC
MPPSET controlled by CE
CHARGE termination is disabled by grounding TERM_EN
For 10A output,
Inductor = 3.3uH
Capacitor = 40uF
Resistor = 4mR
S_VIN
VBAT
GND
CE
S_VIN
VBAT
GND
CE
Title
Size
A3
Number
Revision
Date:
8/31/2023
Sheet of
File:
D:\Github\..\Solar_Charge_Controller.SchDoc
Drawn By: